

# TWO-SWITCH SIMULTANEOUS LED CONTROL SYSTEM

## 1. Aim

To design and build a simple LED control circuit using two switches, an LED, resistor, power supply, and connecting wires, in which the LED glows only when both switches are pressed simultaneously.

## 2. Problem Statement

In some electronic control systems, an output should be activated only when two conditions are satisfied at the same time. To demonstrate this concept, two switches are used to control a single LED. The LED is designed to glow only when both switches are pressed simultaneously.

## 3. Proposed Solution

The proposed system uses two switches to control the operation of an LED. When both switches are pressed at the same time, the circuit is completed and current flows through the LED, causing it to glow. If only one switch is pressed, the required circuit condition is not satisfied and the LED remains OFF. This project demonstrates the basic concept of AND logic using switches.

## 4. Components Required

- LED
- Two Push Buttons / Switches
- Resistor
- Breadboard
- Battery / DC Power Supply
- Jumper Wires

## 5. Circuit Working

The two switches are connected in series with the LED circuit. The resistor is connected in series with the LED to limit the current and protect the LED from excessive current.

### Working condition:

Switch 1 OFF + Switch 2 OFF → LED OFF

Switch 1 ON + Switch 2 OFF → LED OFF

Switch 1 OFF + Switch 2 ON → LED OFF

Switch 1 ON + Switch 2 ON → LED ON

Therefore, the LED glows only when both switches are pressed simultaneously. This operation is similar to the function of an AND gate, where the output becomes HIGH only when both inputs are HIGH.

## 6. Procedure

1. Place the components on the breadboard.
2. Connect the LED with the correct polarity.
3. Connect a suitable resistor in series with the LED.
4. Connect the two switches in series according to the circuit.
5. Connect the positive terminal of the power supply to the circuit.
6. Connect the negative terminal to the circuit ground.
7. Check all the connections carefully.
8. Switch ON the power supply.
9. Press only Switch 1 and observe the LED.

10. Press only Switch 2 and observe the LED.
11. Press both switches simultaneously.
12. Observe that the LED glows when both switches are pressed.

## **7. Applications**

- Safety Systems – Two conditions can be required before activating an output.
- Industrial Control – Two switches can be used for simultaneous machine control.
- Educational Projects – Demonstrates basic AND logic and circuit operation.
- Access Control Systems – Two conditions can be used to activate an output.
- Electronic Control Circuits – Useful for demonstrating multiple-input control.

## **8. Advantages**

- Simple and easy to understand.
- Low-cost circuit.
- Easy to assemble on a breadboard.
- Demonstrates basic logic operation.
- Provides simultaneous two-switch control.
- Suitable for beginners and educational demonstrations.
- Helps understand series connections and current flow.

## **9. Result**

The two-switch LED control circuit was successfully built and tested. The LED remained OFF when only one switch was pressed. When both switches were pressed simultaneously, the LED glowed successfully. Thus, the project demonstrated the basic AND logic operation using two switches and an LED.

## **10. Conclusion**

This project demonstrates how two switches can be used to control a single LED based on simultaneous input conditions. The LED glows only when both switches are pressed at the same time. The project provides a practical understanding of switches, current flow, series connection, LED operation, and basic AND logic.

## **11. Project Demonstration**

Project Video / Demo: [Add Video Link Here]

## 12. Circuit Diagram / Project Setup

The following images show the actual breadboard setup used for the project.

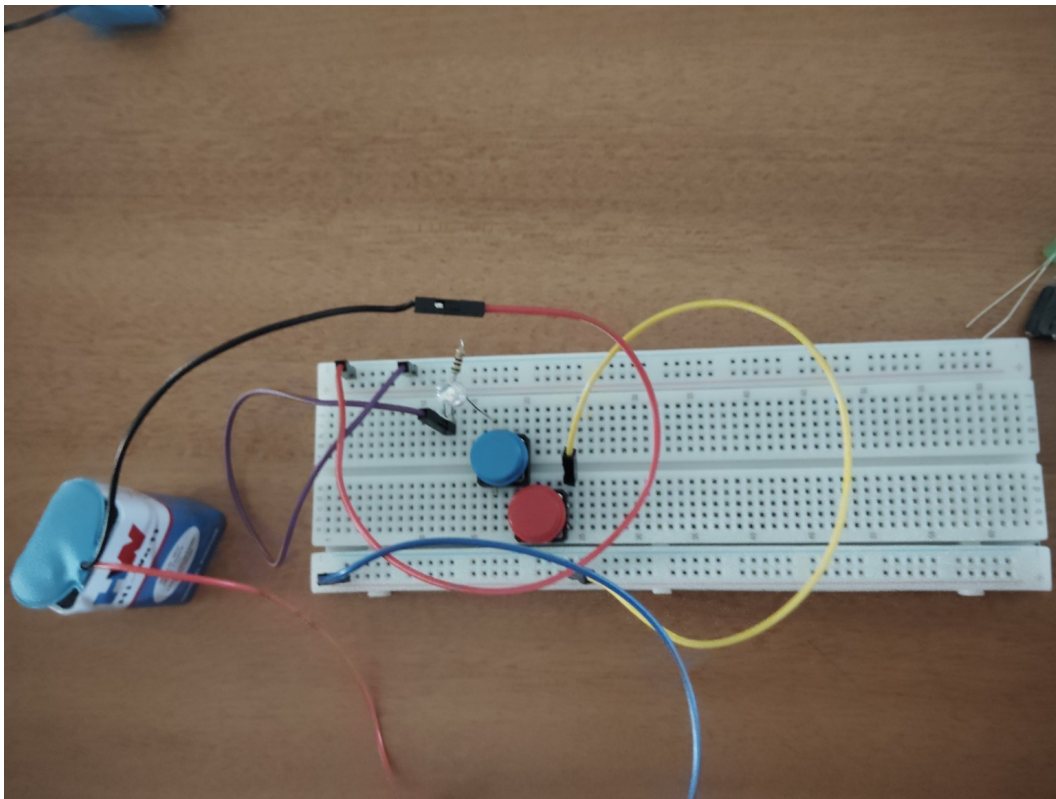


Figure 1: Project circuit setup

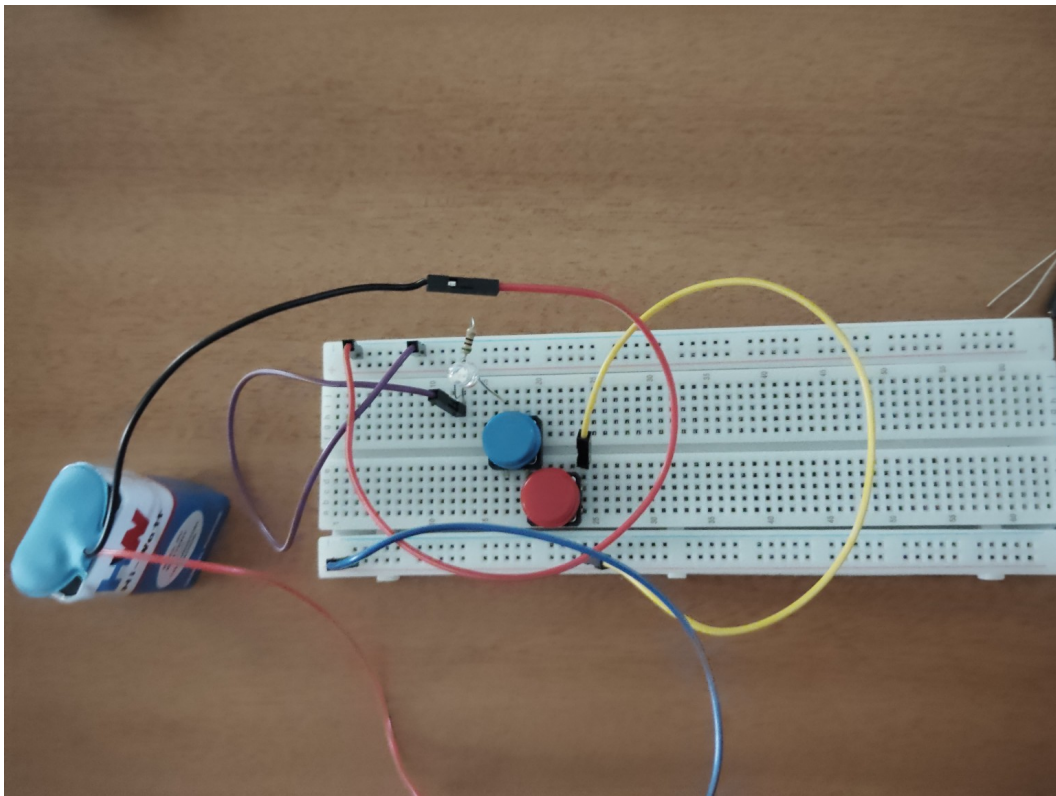


Figure 2: Project circuit setup

## 13. Project Images

The above photographs have been included as the circuit setup and working project images.